Nose Wheel (Mod S.O.O. 6048 and Mod 6/2183) – Maintenance Practices

1. General

The nose wheel is carried on the axle mounted in the fork of the nose gear, and is of the split hub type to facilitate removal and installation of tubeless tires. An O-ring is installed between the two halves of the hub to provide an airtight housing for the tire. The nose wheel is fitted with a 36 x 13.00 x 12 tire. (For STD refer to 32-40-61, Nose Wheel – General Data).

2. Removal/Installation

Refer to Figure 1 for details.

A. Remove Nose Wheel

1) Jack aircraft until nose wheel is clear of ground. (Refer to 07-00-00, Lifting and Shoring – General Data).

2) Remove and retain nuts, clamps, washers and bolts securing axle to fork.

3) Remove and retain the two lock screws locking retaining nut to adjustment spacer.

4) Hold adjustment spacer with wrench, and slacken retaining nut.

5) Support nose wheel and withdraw axle from fork, spacers and nose wheel. Lower nose wheel and retain spacers.

6) Remove snap ring bearing seals and cone bearings from inboard side, snap ring shim outboard grease bearing, spacer and bearing from nose wheel.

7) Suitably protect nose wheel bearings from contamination.

B. Install Nose Wheel

1) Ensure nose wheel bearing housings are clean, then install snap ring cone bearings, shim, spacer, bearing seals and snap ring, as per Para 2.A.6).

2) Install nose wheel discs with plug opposite valve.

3) Lightly lubricate axle with grease (MIL–PRF–81322), support nose wheel in fork, position spacer, and adjustment spacer with retaining nut, align locating bolt holes with holes in fork and insert axle through fork, wheel and spacers.

Note Ensure that adjustment spacer is installed so that retaining nut is adjacent to left leg of fork.

4) Install axle locating bolts, clamps, washers and nuts. Use washers to a maximum thickness of 0.130 inch between clamp and fork hub as required to mate flats. Coat bolts with corrosion preventive compound MIL–PRF–16173 Grade 2 (Shell CG929) prior to installation.

5) Extend adjustable spacer until wheel bearings begin to bind, then back off the next screw locating slot. Insert lock screws and lockwire.

6) Lower aircraft to ground.

7) Check tire for correct inflation. (Refer to 12-10-20, Landing Gear – General Data).

Note If installation of nose wheel on aircraft is part of gear configuration change then do step as per Para 2.B.8).

8) Do procedure for gear configuration change as per 32-10-11, Main Gear Leg – General Data, Para 2.C.

C. Remove Nose Wheel Tire (with tube)

1) Remove wheel/tire assembly from aircraft. (Refer to Para 2.A).

2) Release all pressure from tire by slackening air valve core.

3) Remove the valve stem bushing from wheel assembly and valve core.

4) Remove valve core completely from wheel assembly.

5) Using tire demounters, wooden wedges, or an approved tire machine, separate edges of tire from wheel hub beading.

6) Remove bolts, washers and self-locking nuts, holding hub halves, and remove O-ring and tire.

7) Carefully separate hub halves and remove o-ring and tire and tube. Check O-ring for serviceability and discard if damaged.

D. Install Nose Wheel Tire (with tube)

1) Check hub for scratches, scores and cleanliness. Clean tire area of hub with a dry cleaning solvent. Dust area with french chalk (talc).

2) Partially inflate tube and position in tire. Align yellow stripes on the tube base with red dot on the tire side wall.

3) Position hub half containing air valve on a bench. Place serviceable tire and O-ring on hub, with red spot in tire aligned with air valve. Position valve stem through valve hole and secure with valve stem bushing. Ensure valve stem bushing is snug.

4) Lubricate threads of bolts and bearing surfaces of nuts, bolt heads, and washers, with an anti-seize compound (MIL–T–5544A).

5) Ensure O-ring is in position and install other half of hub.

Note Ensure that the “L” stamping (which indicates the light side) on the wheel halves are aligned opposite approximately 180 degrees apart.

6) Install bolts, washers and self-locking nuts, to secure hub halves and tighten bolts to torque value stamped on wheel hub.

Note Ensure NAS1149F0663P washer is installed under counter sunk washer (MS2002C6) as per Figure 1.

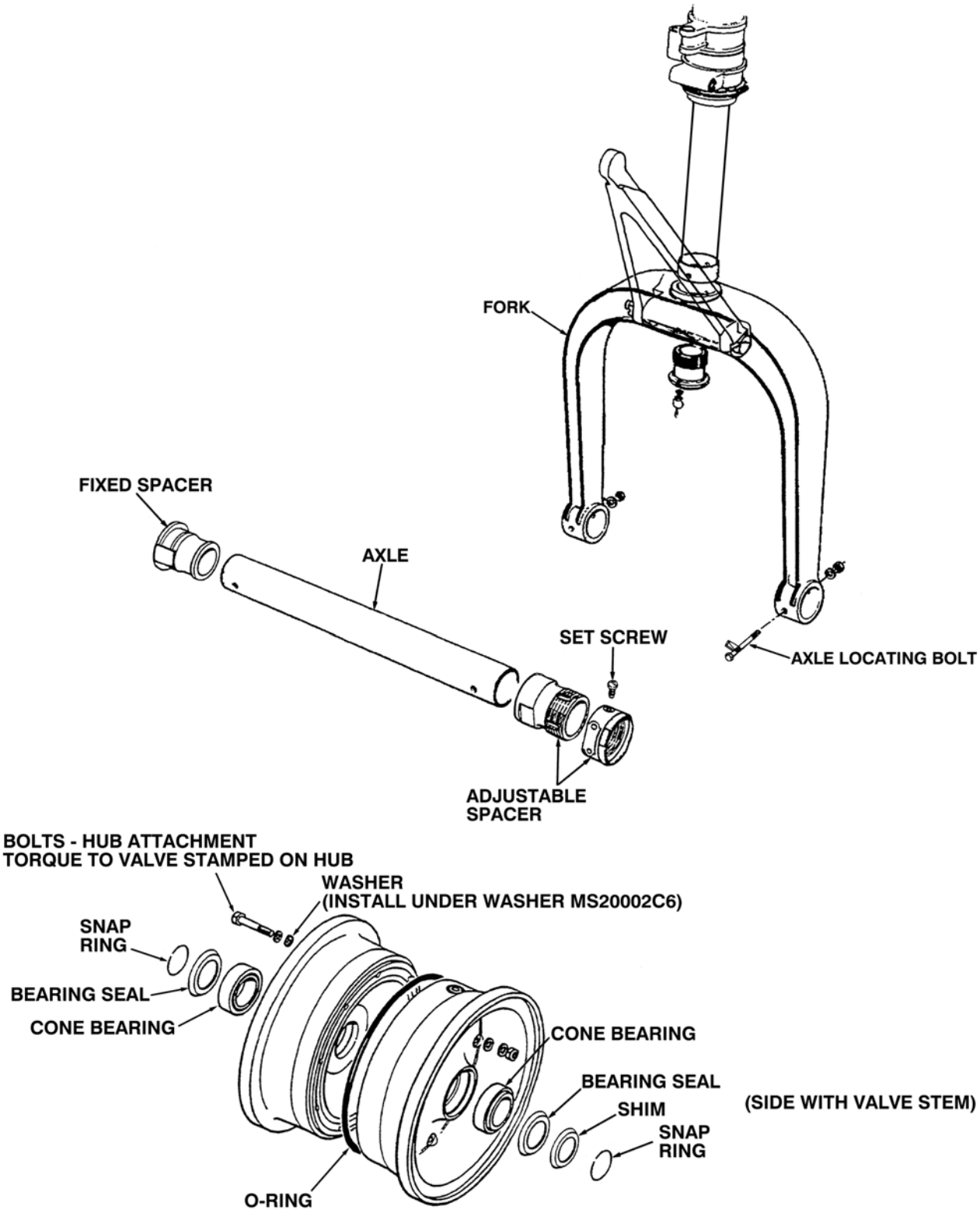
WARNING DO NOT INFLATE TIRE UNTIL ALL BOLTS HAVE BEEN PROPERLY TIGHTENED. IT IS RECOMMENDED THAT ALL MOUNTED TIRES BE INFLATED INSIDE A METAL CAGE TO AVOID POSSIBLE INJURY TO PERSONNEL.

7) Inflate tire to correct pressure. (Refer to 12-10-20, Landing Gear – General Data).

Note It is recommended that wheel balance is checked when a new tire is installed on hub. The following method of balancing is recommended: check static balance, and add adhesive strip weight as necessary, applying 1/2 of required weight to each side of the hub.

8) Apply a 1” x 3” creep mark using a suitable contrasting coloured paint between tire and rim.

9) Install wheel on aircraft.



Nose Wheel Installation (Mod S.O.O. 6048 and Mod 6/2183)
Figure 1

3. Inspection/Check

A. Inspect Nose Wheel Hub (Mod S.O.O. 6048 and Mod 6/2183)

1) Remove nose wheel and nose wheel tire (refer to Para 2.A and Para 2.C).

2) Clean hub halves with dry cleaning solvent (P–S–661 or equivalent) and dry with compressed air or a lint-free cloth.

3) Inspect hub halves for cracks and breaks using the fluorescent penetrant or an approved dye penetrant method, noting in particular the bead seat, tubewell, and web at junction areas. Replace hub if cracks or chipping of rims is evident.

Note Scratches, nicks and other surface blemishes on hub halves can be removed by sanding with emery cloth removing as little metal as possible. Polish the repaired surface and repaint the repaired areas. Use one coat of Chromated Epoxy Primer to MIL–PRF–23377 on packing groove and mating surfaces of hub halves. Use two coats of Chromated Epoxy Primer to MIL–PRF–23377 and one finish coat of aluminum lacquer (MIL–L–7178) for other areas.

4) Inspect bearing cups and replace if damaged.

5) Install tire to nose wheel and install nose wheel (refer to Para 2.B and Para 2.D).